Urban Building Classification (UBC) V2—A Benchmark for Global Building Detection and Fine-Grained Classification From Satellite Imagery

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***Abstract*— Datasets play a key role in developing superior building detection approaches. However, most of the previous work focuses on accurate building masks and scale expansion, while the categories are always missing, which hinders the further analysis of urban development and cultures. Therefore, we propose a benchmark for building detection and fine-grained classification from very high-resolution (VHR) satellite imagery. An extensive annotation is performed for about 0.5 million building instances with 12 fine-grained roof types and individual polygons. The annotation of building functions of two cities in the previous version (UBCv1) (Huang et al., 2022) is also integrated. To ensure the building variety, it consists of VHR optical images of 20 unique cities worldwide with various landforms and styles of architecture. Its variety and fine-grained categories pose great challenges and meanwhile provide a foundation for the building extraction and fine-grained classification on a global scale. Besides, 17 cities are provided with finely aligned synthetic aperture radar (SAR) images, which can be used for the development and evaluation of approaches optionally based on optical, SAR, or multimodal images. Significantly, the proposed benchmark is used as the base of the 2023 IEEE GRSS Data Fusion Contest (Persello et al., 2023). The dataset and codes of the baseline methods are available at: https://github.com/AICyberTeam/UBC-dataset/tree/UBCv2.**

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***Index Terms*— Building detection, fine-grained classification, instance segmentation, multimodal, optical imagery, synthetic aperture radar (SAR).**

I. Introduction

BUILDINGS are an important urban infrastructure and play an essential role in human habitation and socioeconomic activities. With the rapid advancement of very high-resolution (VHR) imaging techniques for remote sensing, the detection of buildings from remote sensing images can be achieved in large areas at a low cost. Automatic determination of precise boundaries and fine-grained categories of buildings can be used for emergency management, 3-D reconstruction, map making, urban planning, etc. [3], [4], [5]. A further analysis in the changes in land covers and buildings can help humans analyze urban development over time [6], [7], [8].

Many building datasets have been proposed for the development of building extraction algorithms [9], [10], [11], [12], [13], [14], [15], [16], [17]. Most datasets, however, share the following two limits: 1) buildings are usually regarded as a mask, i.e., a single class, which limits the potential for a further analysis of urban development and cultures; and 2) the building varieties in appearance and spatial arrangement are limited. It prevents the building detection algorithms from generalizing to diverse buildings on a global scale.

First, most building datasets [10], [11], [12] focus on quantity, but lack fine-grained building categories. The popular datasets, such as the SpaceNet series [10], [11], INRIA [14] and TorontoCity [12], provide high-quality building annotations in the form of masks or polygons without categories. Only few datasets [15] contain information on the building categories, which are quite coarse. An important reason for the lack of a large-scale building dataset with fine-grained categories is that there is no unified standard or consensus on how to classify buildings. Another reason is that manually labeling buildings, especially in terms of fine-grained categories, is very difficult from remote sensing images. We present a remote sensing dataset for building detection and fine-grained classification, in which buildings are categorized into 12 fine-grained classes based on their roof types, such as gable roofs, gambrel roofs, and arched roofs. The roof categories are carefully defined referring to OpenStreetMap (OSM) [18],

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TABLE I
COMPARISON OF BUILDING DATASETS

Datasets	Source	Buildings	Images	Tile size	cities	Categories	Image format	Fine-grained	Modality	Resolution	Task
SpaceNet 2 [10]	WorldView 3	302,701	24,586	650	4	1	tif	N	RGB, MSI	0.3 m	SS/IS
SpaceNet 6 [11]	WorldView 2, Cappela36	48,000	20,408	900	1	1	tif	N	RGB, NIR, SAR	0.5 m	SS/IS
Toronto City [12]	Drone/Vehicle cameras	400,000	56,000	5,000-10,000	1	1	tif	N	RGB, NIR	0.05-0.1 m	SS/IS
GFB [13]	GaoFen-3, Google Earth	-	13,185	256/512	9	1	-	N	RGB, SAR	1 m	SS
INRIA [14]	USGS	-	360	5,000	10	1	tif	N	RGB	0.3 m	SS
DSTL [15]	WorldView 3	8,045	450	3,350	3	5	tif	Y	RGB, MSI	0.3 m	SS/IS
SemCity Toulouse [16]	WorldView 2	9,455	16	3,452	1	1	tif	Y	PAN, MSI	0.5 m	SS/IS
JHU dataset [17]	Vricon	10,300	300	2,000	2	1	-	N	RGB, DSM	0.5 m	SS/IS
UBCv1 [1]	SuperView, GaoFen-2	41,586	800	600	2	61	tif	Y	RGB	0.5-0.8 m	SS/IS
UBCv2	SuperView, GaoFen-2, GaoFen-3	499,339	11,336	512	20	12	tif	Y	RGB, SAR	0.5-0.8 m	SS/IS

CityGML [19], and Google Maps, as well as considering the characteristics of remote sensing data. The annotation quality is ensured by a strict quality control mechanism in the annotation pipeline.

Note that our previous work, i.e., UBCv1 [1], comprised a unique classification of building functions (use/occupation) for data from Beijing and Munich. In this extended version, we do not provide annotations for building functions for the 18 other cities, because of the large amount of manual work and the high cost. We, however, keep the existing annotations for the two cities to enable research on the classification of functions of buildings in urban areas.

Second, the building diversity is not given special attention in major building datasets [10], [11], [12], [15], [16], [17]. We present the number of cities in popular building datasets in Table I, which reveals that they usually only cover a limited extent of urban areas, resulting in low diversity. On one hand, diversity is reflected in the appearance. As presented in Fig. 1, Beijing's buildings usually present a modern concrete style and a neat systematic layout, while buildings in Munich are much lower and more crowded. Instead, Barcelona has a regular distribution of small blocks as the basic elements in the neighborhood while the selected buildings in Jacksonville are characterized by evenly distributed villas. To ensure diversity in terms of building appearance, we collect VHR remote sensing images with unique cultural characteristics for 20 cities around the world, such as Beijing, Munich, Jacksonville, and Barcelona presented in Fig. 1.

On the other hand, diversity is reflected in the spatial arrangement of buildings. Most datasets [12], [13], [15] have few areas with dense building clusters, since it is very difficult to label the closely adjacent and crowded building instances. However, dense clusters of buildings are common in practice. Therefore, finely segmenting individual building instances is a challenge that cannot be ignored. The UBCv2 contains crowded buildings. As presented in Fig. 1, the buildings

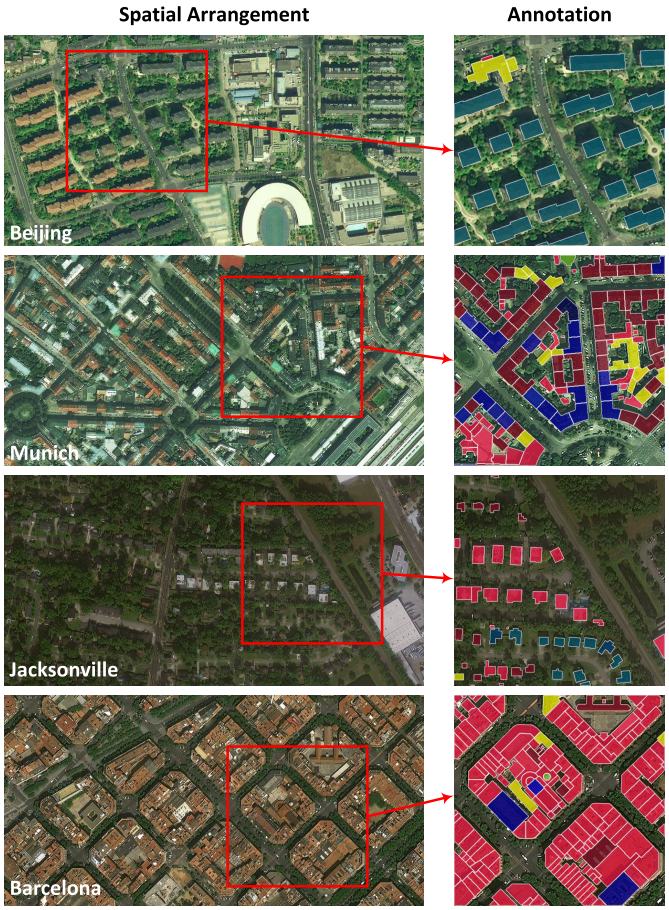


Fig. 1. (Left) Four selected urban areas with typical architectural characteristics. (Right) Annotation details in the image tiles in the UBCv2 dataset. Colors indicate various classes of roof types. Buildings in different cities show a large diversity of appearance and spatial arrangement.

in Barcelona show an extremely dense distribution. Besides Barcelona, seven of the 20 cities include such a dense distribution and the labeling of individual buildings is carefully

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referenced to high-quality data sources such as OSM [18] and Google Maps.

Furthermore, we provide multimodal data including RGB and synthetic aperture radar (SAR) images in 17 cities. We have finely aligned the 1-m resolution spotlight SAR images from GaoFen-3 satellites with the optical images. Optical images can capture colors (spectrums) and textures of buildings, which are intuitive to human eyes. Complementally, SAR is a kind of range detector and encodes more explicit range and geometry information. Besides, SAR is immune to weather and lighting conditions and can observe the ground around the clock. Therefore, the joint usage of the two modalities is expected to benefit the interpretation of satellite images intuitively.

This benchmark is used as the base of 2023 IEEE GRSS Data Fusion Contest: Large-Scale Fine-Grained Building Classification for Semantic Urban Reconstruction (DFC 2023) [2]. The DFC 2023 provides finely aligned optical and SAR image pairs for 17 cities, and the UBCv2 contains additional optical data for three new cities (Beijing, Hefei, and Munich).

The main contributions of the proposed work can be summarized as follows.

1) *Fine-Grained Categories for Individual Building Instances:* We define 12 fine-grained roof categories for buildings taking into consideration the widely used standards (OSM, CityGML, and Google Maps) and the characteristics from a satellite perspective. Each building instance is carefully labeled with a polygon and category.
2) *Rich Diversity From Global Coverage:* This dataset contains urban areas of 20 unique cities from five continents, so as to provide building varieties in terms of both appearance and spatial arrangement.
3) *Multimodal Data With Large-Scale Annotations:* The dataset provides finely aligned optical and SAR image pairs for 17 cities and 350 866 precisely labeled building instances.
4) *Improved Baselines With Explicit Geometry Modeling:* We conduct experiments with representative approaches on UBCv2 and improve the performance via introducing a classwise geometric structure feature enhancement module. The results can be used as the baselines and give inspiration to future work.

II. Related Work

In this section, we first introduce the large-scale object detection datasets for remote sensing in Section II-A. Related building datasets are listed in Section II-B, and a few datasets for building classification are given in Section II-C. We investigate the current multimodal building datasets containing SAR images in Section II-D. Section II-E summarizes approaches focusing on building detection. And approaches for fine-grained classification are introduced in Section II-F. A detailed comparison for various quantitative indicators of our dataset with other datasets containing buildings is given in Table I.

A. Large-Scale Remote Sensing Object Detection Datasets

In the era of data-driven deep learning, large-scale datasets are essential for the development of approaches for the interpretation of remote sensing images. Therefore, many datasets have been made available focusing on different aspects. For example, the DOTA dataset [20] has become the baseline dataset in remote sensing for the validation of object detection algorithms. It contains more than 170 million objects. The iSAID dataset [21] is introduced based on the DOTA dataset but comprises additional annotations in the form of instance masks. The FAIR1M dataset [22] collects 15 266 high-resolution remote sensing images and provides more than one million objects with fine-grained labels in five categories and 37 subcategories. All these datasets have substantially contributed to the development of remote sensing object detection algorithms. Yet, our dataset particularly focuses on buildings and comprises 0.5 million building instances.

B. Building Datasets

In urban areas, where buildings are often the defining part, building detection has attracted the attention of many researchers. Current building datasets are mainly annotated in the form of polygons or pixels. We first present datasets annotated using polygons. The SpaceNet 2 Challenge [10] comprises about 0.3 million building footprints in 24 586 tiles of size 650 × 650 from images of the four cities Las Vegas, Paris, Shanghai, and Khartoum with a ground sample distance (GSD) of 0.3 m. The Toronto City dataset [12] makes available aerial imagery from the Toronto area with a GSD of 0.05–0.1 m containing about 400 000 buildings. The JHU dataset [17] provides both 0.5-m DSMs and 0-5 m RGB orthorectified images. The SemCity Toulouse benchmark [16] provides 9455 building annotations in the 0.5-m multispectral images. Unlike the above datasets using polygons, the INRIA [14] dataset labels buildings in pixel format. It consists of 0.3-m orthorectified images registered with precise cadastral records from ten cities in the U.S. and Austria.

We choose to annotate buildings in the form of polygons. The advantage of annotation by means of polygons is that it is easy to produce a format for instance segmentation. Our dataset specifically focuses on the detection of building roofs, because building roofs are the main parts of buildings when seen from above. And the annotation of roofs can avoid inaccurate labels since the view of footprints might be blocked by the roof overhang.

C. Building Classification Datasets

Currently, only few datasets distinguish categories of buildings at a relatively coarse-grained level. The DSTL Satellite Imagery Feature Detection dataset [15] strives for coarse-grained building classification, i.e., fortified, residential, nonresidential, and fuel storage facilities. It provides 450 multispectral satellite images in 16 bands and comprises 8045 buildings in a GSD of 30 cm. Instead, our dataset focuses on the fine-grained classification of building roofs and has six times the number of instances compared with DSTL. We have

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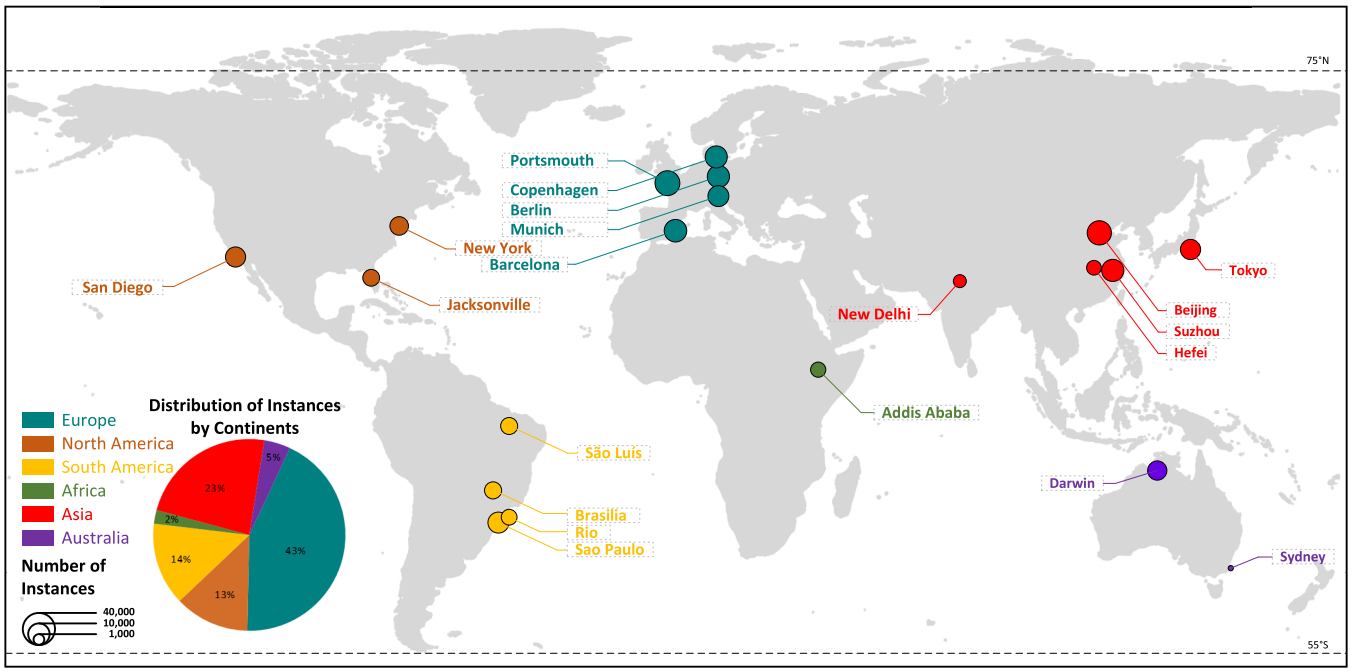


Fig. 2. Global distribution of 20 cities: each circle represents a city, and each continent is marked with a different color. The size of the circle indicates the number of building instances. The pie chart gives the distribution of instances over the five continents.

designed fine-grained categories based on the morphology of building roofs as seen in satellite images. Specifically, 12 fine-grained categories of roof structures have been defined, including flat, gable, gambrel, row, multiple eave, hipped v1, hipped v2, mansard, pyramid, arched, revolved, and other roof.

D. Multimodal Building Extraction Datasets

Since SAR is mostly independent of weather and lighting, SAR satellites can observe the ground around the clock. Compared with unimodal data, multimodal image data from optical and SAR imagery provide more information about surface objects [23]. Therefore, research on the multimodal fusion of aligned RGB and SAR imagery [24], [25], [26], [27] has also become an important topic in the scientific community. SpaceNet 6 [11] is a large-scale multimodal dataset, comprising 20 408 finely aligned RGB and SAR image pairs. It provides about 48 000 building footprints. The GaoFen-3 SAR dataset [13] comprises 302 701 SAR images with a GSD of 1 m for buildings. It covers urban and rural areas in typical cities, e.g., Berlin, Yokosuka, and San Diego. In this article, we additionally provide a multimodal version of the benchmark consisting of finely aligned RGB and SAR image pairs with a larger quantity of building instances.

E. Building Detection Methods

Building detection methods are mainly divided into two categories: semantic segmentation and instance segmentation. Wagner et al. [28] propose to predict building masks and pixels at the contours. The latter are further used to separate the connected masks of adjacent buildings. Prevailing instance segmentation methods can be divided into single- [29], [30], [31] and two-stage [32], [33], [34] approaches. Li et al. [35] propose to perform key point detection on building masks derived by a general two-stage instance segmentation framework. The detected key points are subsequently connected to form closed polygons. Sun et al. [36] introduced auxiliary geographic information system (GIS) data to improve the building extraction of individual instances in SAR images.

F. Fine-Grained Classification

Fine-grained image classification has been a challenging problem for a while [37], [38], [39], [40], [41]. Specifically, subcategories with minor differences within the same category are to be classified. As summarized in [42], the fine-grained classification problem can be solved in two ways: 1) recognition by classification with subnetworks and 2) end-to-end feature encoding. The former [43], [44] learns discriminative representation by detecting the basic components of the object, while the latter [45], [46] improves the classification performance by increasing the model's fine-grained representation ability. Instead of the above image-level fine-grained classification methods, we specifically focus on solving the fine-grained classification of buildings at the instance level.

III. Dataset

A. Area of Interest

To construct a global remote sensing dataset for diverse buildings, we have explored available remote sensing data from 20 cities worldwide. The geographical distribution of the annotated building instances is presented in Fig. 2. The cities reside in five continents, ensuring a diversity of urban building styles. Fig. 3 demonstrates the spatial arrangement and detailed annotations of samples from each city. Every city shows a typical layout that correlated to its unique historical

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Fig. 3. Specific layout demonstration with detailed annotations (white edges) of 20 cities.

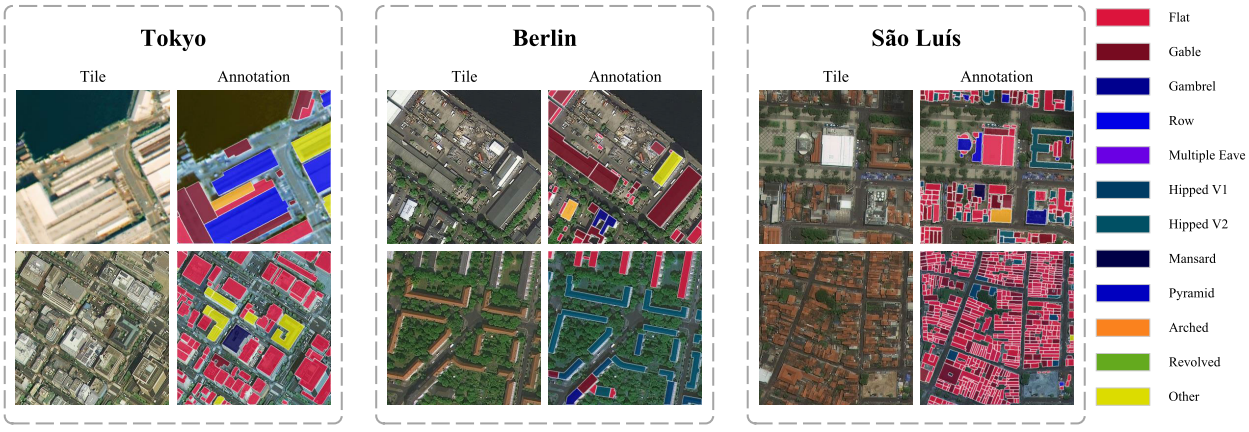


Fig. 4. Various scenes collected from some cities in the UBCv2 dataset, i.e., Tokyo, Berlin, and São Luís.

development and geographical pattern. The buildings in each city exhibit distinctive spatial arrangement, i.e., the dense regular arrangement of small buildings in Portsmouth, the commercial high-rises in Beijing, and the sparsely distributed villas in Jacksonville. For the 20 areas of interest, there is a high diversity within and among geographies. As shown in Figs. 4 and 5, the images collected in UBCv2 are from various scenes such as urban, peri-urban, business districts, industrial, and slum areas. For Tokyo, the roofs of industrial buildings (first row) are often gable and row roofs. But commercial

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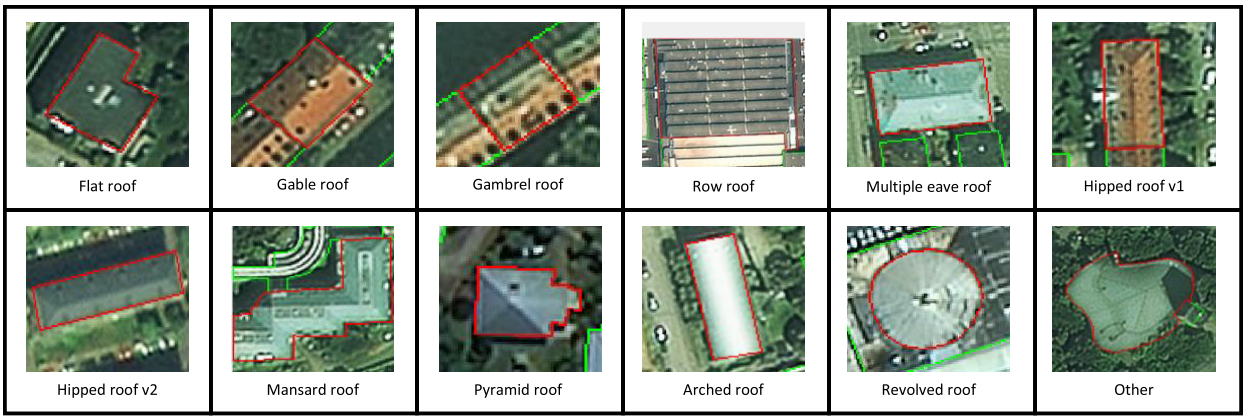


Fig. 5. Examples of roof-type classes in the UBCv2 dataset. The red outline indicates the annotated footprint of the building.

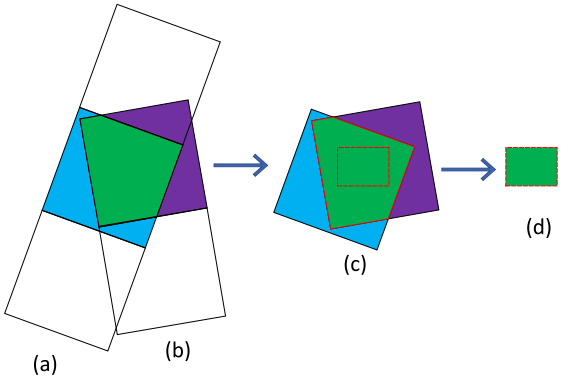


Fig. 6. Workflow of AoI selection. First, cities are determined with overlapping areas in (a) optical and (b) SAR tile sequences. The tiles with the largest intersection area (blue and purple tiles) are selected. (c) and (d) AoI is cropped in the green area by evaluating the image quality together with the completeness of the OSM data.

TABLE II
ACQUISITION TIME AND SAR POLARIZATION FOR ALL CITIES

City	Date (Optical)	Date (SAR)	SAR Polarization
Suzhou	2020-03-25	2021-05-16	HH
Tokyo	2022-02-17	2022-02-13	HH
New Delhi	2017-04-12	2017-09-03	HH
San Diego	2022-02-16	2022-06-07	HH
New York	2019-10-25	2020-05-12	HH
Jacksonville	2019-11-26	2019-10-09	HH
Addis Ababa	2019-03-31	2019-10-20	HH
Sydney	2021-08-03	2021-06-24	HH
Berlin	2020-08-07	2019-11-29	HH
Copenhagen	2020-04-05	2020-04-05	HH
Portsmouth	2022-02-11	2022-03-25	HH
Barcelona	2018-05-21	2018-11-19	HH
Brasilia	2019-11-13	2019-11-03	VV
São Luís	2020-05-03	2022-01-30	HH
Rio	2018-01-17	2019-10-19	VV
Sao Paulo	2020-04-22	2019-10-20	VV
Darwin	2020-07-13	2020-08-26	HH
Munich	2021-07-21	-	-
Hefei	2018-01-10	-	-
Beijing	2021-04-14	-	-

buildings in business districts (second row) are more with flat roofs. For Berlin, many buildings have gable roofs in the peri-urban (first row), but flat and hipped roofs are the main types in urban areas (second row). For the slum areas (second row) in São Luís, low-rise buildings with flat and gable roofs are the dominant.

We have considered the compatibility of images from various sources, i.e., optical and SAR satellites. We looked for cities, where the data are available in both the modalities and the interval between acquisitions does not exceed two years. The date of the optical (GaoFen-2 and SuperView) and SAR (GaoFen-3) images for all the 20 selected cities is given in Table II. Besides, we also provide the polarization of SAR images for 17 cities. Based on the optical and SAR images [Fig. 6(a) and (b)] of an urban area, we first determine their intersection part (green), from which the area of interest (AoI) is cropped based on a comprehensive evaluation of both the image quality and the completeness of the available OSM [18] data (dashed red rectangle in Fig. 6).

B. Image Collection and Preprocessing

Multiple sensors with multiple resolutions, specifically the Chinese GaoFen-2, SuperView, and GaoFen-3 satellites, have been chosen. SuperView-1 (or "GaoJing-1" in Chinese) is a constellation of commercial VHR earth observation satellites. The data collected by SuperView-1 include panchromatic (0.5-m GSD) and multispectral (2-m GSD) images. It has a maximum imaging range of 60 × 70 km. The GaoFen-2 satellites from Chinese National Space Administration (CNSA) include panchromatic and multispectral imagery with a GSD of 81 and 324 cm, respectively. The maximum imaging range of GaoFen-2 satellites is 45 km. The GaoFen-3 is a constellation of civilian SAR imaging satellites which produce C-band polarimetric SAR images. The spotlight mode of the GaoFen-3 satellite produces single-polarized SAR images with a GSD of 1 m.

The RGB and panchromatic data from GaoFen-2 and SuperView are pan-sharpened to derive high-resolution RGB imagery with a GSD of 0.5 and 0.8 m, respectively. Note that the GaoFen-2 images are then upsampled to 0.5-m resolution to preserve the spatial consistency of all the image tiles. Optical and SAR images are finely aligned and projected to the Universal Transverse Mercator (UTM) coordinate system.

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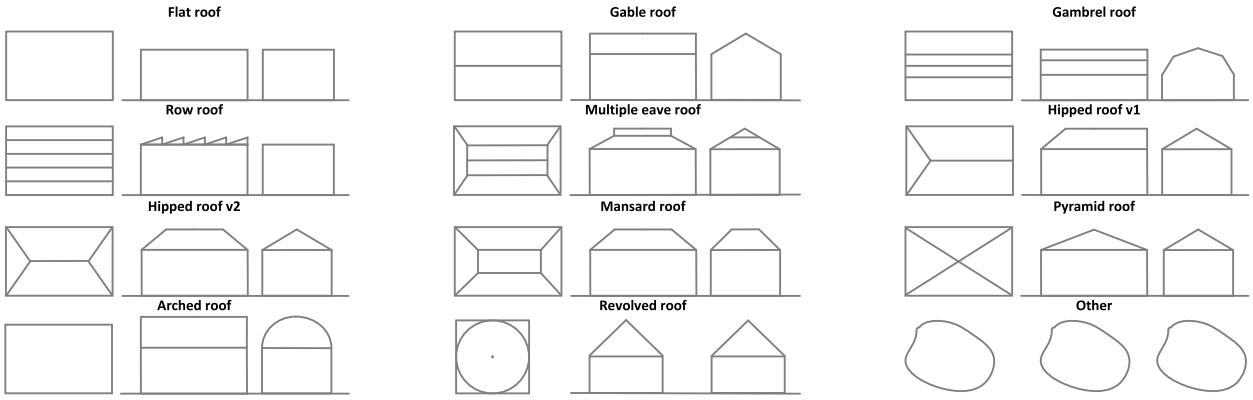


Fig. 7. Geometrical information of roof classes in the UBCv2 dataset. Three views (top, front, and right) of a typical example are shown for each class.

C. Category Design

We define 12 roof classes, i.e., flat, gable, gambrel, row, multiple eave, hipped v1, hipped v2, mansard, pyramid, arched, revolved, and other roof. Fig. 5 presents the roof classes in the form of examples, and the definition with three views is given in Fig. 7. To define useful classes that are compatible with shared insights and with our available data, a concise definition is first summarized according to the categories of the prevailing data sources, i.e., OSM [18], Google Maps, and CityGML (partially open) [19]. Only then we set up the final 12 classes considering the characteristics and limitations of available data. This makes sure that we can quickly map the building attributes in these data sources to the proposed categories.

In our previous work [1], the UBCv1, we have explored very fine categories for both roof types (9 coarse and 25 fine classes) and building functions (5 coarse and 36 fine classes). The annotation and quality control, especially for the building functions, have proven to be very time-consuming and expensive. Since the number of building instances is significantly extended in UBCv2, we focus on the fine-grained roof-type classification. It is optimized based on our experience learned with UBCv1 and the variety of buildings in the new data. The coarse- and fine-grained classes in UBCv1 are merged into 12 classes considering the practical feasibility of remote sensing data.

D. Annotation

With regard to quality control, experienced experts inspect the preliminary annotations from professional annotators in a strict workflow, where 15 annotators are trained by two inspectors to learn a uniform annotation standard, first. To improve annotation efficiency and reliability, especially for densely adjacent buildings, data from OSM [18] or Google Maps are used if available. The extracted roofs are manually refined using the input images and mapped into predefined classes. Then, the inspectors carry out a thorough check after the annotation results have been produced. Specifically, multiple annotators and inspectors jointly decide on the final labels for controversial results. Finally, a thorough check is conducted on randomly sampled annotations and the final production is completed until the check passes.



Fig. 8. Annotation example: (a) input image and (b) building polygons. The interior polygons of buildings with inner courtyards are also carefully annotated.

The roofs of buildings are annotated using polygons. As shown in Fig. 8, every instance is annotated using polygon format (white contours), and the fill color denotes roof type. It can be seen that our annotation provides the detailed structure of buildings. For the interior structures, buildings with courtyards are carefully annotated.

The distribution of instances over classes is shown in Fig. 9. The value of the vertical axis represents the number of instances for each category. The flat roof type is the most frequent in the dataset. Despite the long-tail distribution among the 12 classes, we carefully balance the number of instances of different categories in the three sets, i.e., training, validation, and test sets.

Instances in UBCv2 show a large variety with respect to areas, which presents a challenge for multiscale building detection. The box plot of the area distribution of the individual classes in UBCv2 is presented in Fig. 10(a). The value of the vertical axis represents the area of instances for each category. Objects with an area of fewer than ten pixels have been excluded. The area of instances in the same class or of different classes varies considerably, with the area span between Q1 (first quartile) and Q3 (third quartile) close to an order of magnitude, e.g., the row, arched, revolved, and flat roofs. For the buildings with row roofs, its Q1 is above 1000. However, the Q3 of pyramid roofs is lower than 1000.

Instances with large aspect ratios pose a great challenge for slender building detection. The aspect ratio means the ratio

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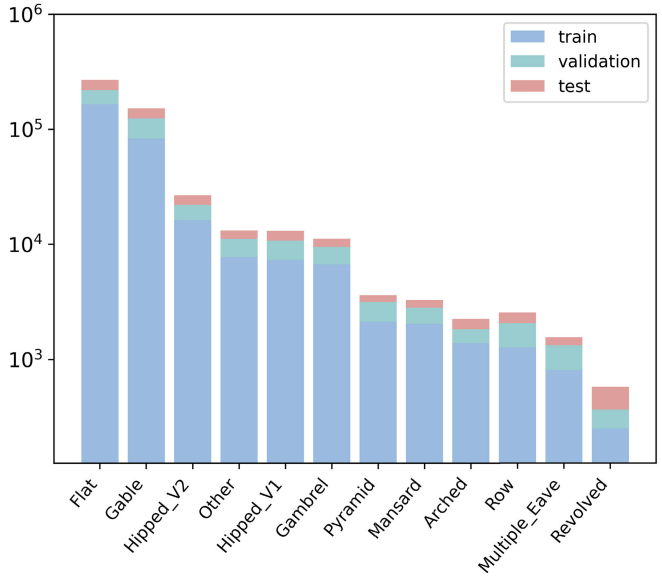


Fig. 9. Statistics of classes in UBCv2.

of the longer side to the shorter one of the bounding box for each building instance. Specifically, given a bounding box with height H_b and width W_b, the aspect ratio is calculated as $\max\{H_b, W_b\}/\min\{H_b, W_b\}$. The box plot in Fig. 10(b) shows the aspect ratio distribution of buildings. The value of the vertical axis represents the aspect ratio of instances for each category. For flat roofs, the most significant aspect ratio reaches 189. The highest mean value for aspect ratio is for arched roofs, while the lowest is for revolved roofs.

E. Multimodal Data Sources

For UBCv2, SAR images with 1-m resolution from the GaoFen-3 satellite are available for 17 cities, as listed in Table II. This provides multimodal data for building detection. The SAR images are combined with the optical RGB images based on a fine geometric alignment. An example of two aligned images is presented in Fig. 11

The SAR images are given in 32-bit float data which makes a quantitative analysis possible. Since the resolution of SAR images is 1.0 m, we upsample them to 0.5 m to obtain a consistent resolution with the RGB images. Considering the absence of SAR data for three cities, our dataset is therefore published in two forms: 1) RGB and 2) RGB + SAR data. RGB unimodal has three more cities than the multimodal form.

F. Construction of the Image Patches

The datasets we provide follow the COCO [47] formatting standards. We first cut all the large image tiles except those with a width/height of fewer than 2000 pixels into medium tiles of 2000 × 2000 pixels without overlap. For each city, we first randomly partition the medium tiles into training, validation, and test sets at a ratio of 6:2:2. Only then each tile is cropped into patches of 512 × 512 pixels with an overlap of 200 pixels. Overall, we produced 11 336 image tiles, with 6556, 2507, and 2273 patches for training, validation, and test sets, respectively. There are 499 339 building instances.

For the multimodal image pairs available for the 17 cities, we keep the same partition for these cities. We calculate the coordinates in the SAR image corresponding to the optical patch based on the geographic information available for each image. The SAR patches are cropped using the corresponding coordinates and resized to the optical patch size of 512 × 512 pixels. We produce 7314 patches in total, with 4008, 1681, and 1625 images for training, validation, and test. The multimodal dataset comprises 350 866 building instances.

IV. Algorithm Design

In this section, we first discuss the general two-stage instance segmentation framework in Section IV-A. We introduce the geometric feature learning for buildings and the transformer structure in Sections IV-B and IV-C, respectively. In Section IV-D, we propose class-wise geometric transformer (CGT) module consisting of a geometric encoder and a class-wise transformer decoder for the fine-grained classification of buildings.

A. General Two-Stage Instance Segmentation Framework

As shown in Fig. 12, backbone network [48], [49], RPN [50], bounding box head [50], and mask head [32] are four basic modules of general two-stage instance segmentation algorithms such as Mask R-CNN [32]. The input image is first encoded by a backbone network, e.g., ResNet [48], and a multiscale feature map is obtained by the feature pyramid network (FPN). Then, RPN extracts a certain number of proposals for which RoIAlign [32] extracts object-level features from the corresponding feature regions. Finally, these object-level features are fed into two parallel branches: 1) the bounding box branch that outputs coordinates and the corresponding classification score for each bounding box and 2) the mask branch that predicts one mask per bounding box.

B. Geometrical Feature Learning for Buildings

From the perspective of satellites, building roofs often exhibit a very distinct geometric structure. As demonstrated in Fig. 7, the gable roof, for example, has a distinct line in the middle of the rectangle, which divides the roof into two similar subrectangles. The hipped roof v2 comprises two symmetrical triangles and trapezoids. In this regard, it is reasonable to assume that these discriminative features in building roofs could be captured in remote sensing images from the top perspective. Therefore, geometric features such as lines, edges, and shapes give promising potential for the classification of building roofs. Heuristically, we enhance a model's ability to represent fine-grained features of buildings via guiding the model to learn geometric features.

Image gradient filters such as the Sobel and Prewitt operator can easily extract the basic contour in an image. In [51], the Sobel operator is used to extract the coarse contour of the building. Then, several circular convolutional layers learn the fine contour of the building. However, it does not use these geometric features in the classification of buildings. In the remote sensing images, building roofs have a distinct geometric structure. These structural features are useful for

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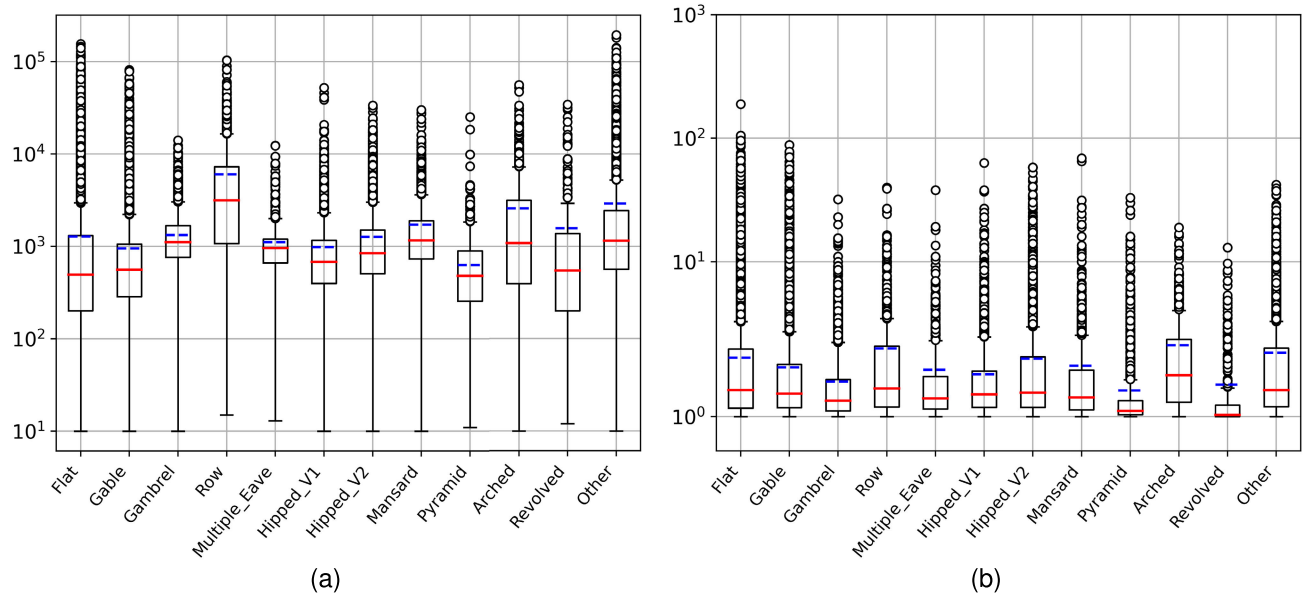


Fig. 10. Distribution of building instances concerning (a) area of footprints and (b) aspect ratio. The red and blue lines indicate the median and mean, respectively. The value of the vertical axis represents the corresponding value of instances for each category, respectively. The upper and lower borders of boxes indicate the upper and lower quartiles, respectively. Circles denote outliers.

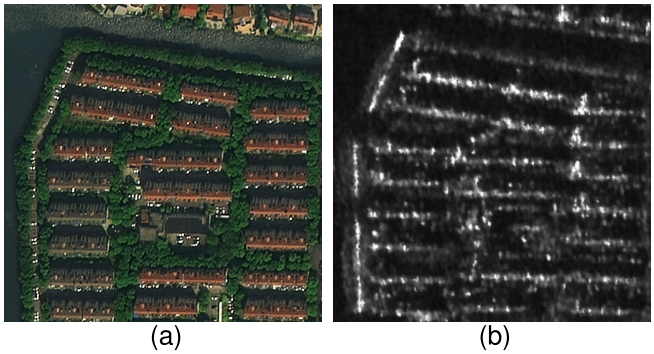


Fig. 11. Example of a multimodal image pair. (a) RGB and (b) SAR images are finely registered.

classifiers to learn fine-grained information about building roofs. Therefore, geometric information such as edges and contours provide auxiliary information that is useful for the roof classification task. We use these geometric features to improve the classification performance of models. And a classwise transformer is designed to implicitly decode the geometric features.

We design a geometric encoder with horizontal and vertical Sobel operators followed by several convolutional layers to guide the model to learn more geometric features, e.g., corners, lines, and contours. Then, those geometric features are flattened as sequential input of a classwise transformer decoder. The CGT module consisting of a geometric encoder and a classwise transformer decoder is proposed to learn the geometric features of buildings in an end-to-end way. We first introduce the basic structure of the transformer in Section IV-C and then describe the details of our CGT module in Section IV-D.

C. Transformer

The transformer was proposed in [52]. It has led to the development of language systems in natural language processing and has recently also demonstrated superior performance in image processing. As shown in Fig. 13, the transformer is a sequential model based on the self-attention mechanism. Its basic block is composed of a multihead attention layer and a feedforward network (FFN). The sequential input tokens are first added with positional encodings and then fed into the encoder which comprises N stacked basic blocks. The input of the decoder is the output of the last layer of the encoder and sequential queries (usually a learnable embedding sequence). The basic structure of the decoder is the same as that of the encoder, and the final outputs are sequentially encoded features with the same length as the query sequence. Fig. 13 shows a transformer with four tokens as inputs and three queries.

D. Classwise Geometric Transformer

Our CGT module guides the model to learn richer geometric features at the object level. It consists of a geometric encoder and a classwise transformer decoder. Specifically, as shown in Fig. 14, the object-level features obtained by RoIAlign are denoted as $F_o \in \mathbb{R}^{D\times H\times W}$. The geometric encoder extracts the structural features $F_h \in \mathbb{R}^{D\times H\times W}$ and $F_v \in \mathbb{R}^{D\times H\times W}$ using the Sobel operator in the horizontal and vertical directions, respectively. Then, F_h and F_v are concatenated in the channel dimension to obtain the structural feature $F_s \in \mathbb{R}^{2D\times H\times W}$, which is transformed by a three-layer CNN obtaining the feature $F_c \in \mathbb{R}^{D\times H\times W}$. This transforms the structural features into the sequential representation space as the input of the transformer decoder.

We add 2-D positional encodings to the geometric feature F_c. They are formulated as follows:

$$\begin{cases} PE(x, y, 2i) = \sin(x/10000^{4i/D}) \\ PE(x, y, 2i+1) = \cos(x/10000^{4i/D}) \\ PE(x, y, 2j + C/2) = \sin(y/10000^{4j/D}) \\ PE(x, y, 2j + C/2 + 1) = \cos(y/10000^{4j/D}) \end{cases} \tag{1}$$

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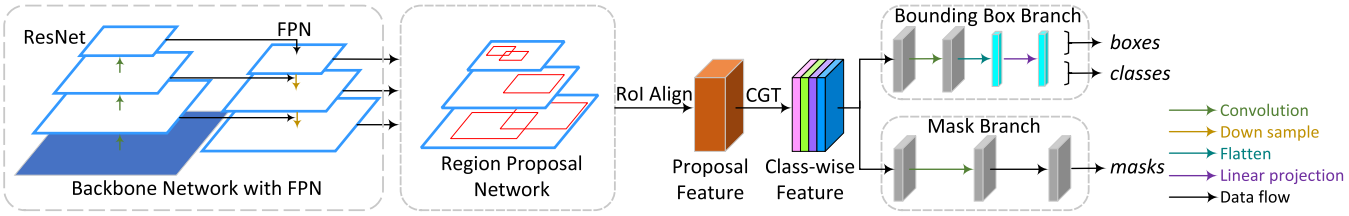


Fig. 12. Framework of classic two-stage instance segmentation methods coupled with the CGT module.

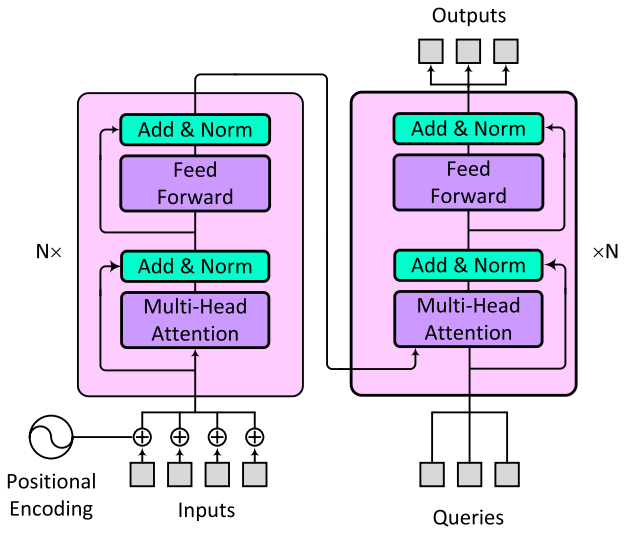


Fig. 13. Basic architecture of the transformer.

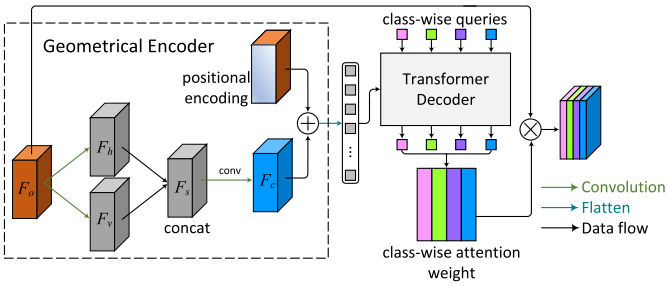


Fig. 14. CGT module.

where (x, y) denotes a point on the geometric feature F_c in the spatial dimension, with x, y in $[0, W)$ and $[0, H)$, respectively. i and j denote indexes in $[0, D/4)$ in the channel dimension. Then, the geometrically positionally encoded features F_g are fed to the transformer decoder. We use a classwise transformer decoder for the geometrically encoded features. The length of the input queries is designed as the number of classes. The queries are a set of parameters that are updated during backpropagation. This classwise design enables the queries to learn classification features and the output can be regarded as classwise attention tokens.

Specifically, the encoded features F_g are flattened as sequential tokens $F_{\text{flat}} \in \mathbb{R}^{D\times HW}$ in the spatial dimension. After transformer decoding, the input tokens are transformed into classwise attention tokens of a fixed length with rich semantic information noted as $T_c \in \mathbb{R}^{D\times(C+1)}$, where C is the number of classes (the background is considered as an additional class).

The original object-level features F_o are transformed to sequential space as $F'_o \in \mathbb{R}^{D\times WH}$. Then, the classwise represented feature F_{cls} is obtained as follows:

$$F_{\text{cls}} = F_o'^{\top} T_c \tag{2}$$

where the transpose and multiplication operations are both performed on the last two dimensions of the features. Finally, F_{cls} is transformed back to the spatial dimension as $F'_{\text{cls}} \in \mathbb{R}^{(C+1)\times H\times W}$.

Our CGT module encodes the classwise information in the channel dimension and transforms the number of input channels from D to $C+1$. In general, $C+1$ is much smaller than D, so the output feature F'_{cls} represents higher level semantic information in a smaller representation space.

V. Experiments

This section introduces the evaluation metrics of the UBCv2 benchmark. The experiments are performed on the UBCv2 dataset with the state-of-the-art models and the proposed CGT module. The results demonstrate the suitability of the dataset and the contribution of the CGT module to the classification task.

A. Task Definition and Evaluation Metrics

In this article, detection and fine-grained classification of buildings are treated as the instance segmentation task. Instance segmentation is a detection task that requires an algorithm to predict a pixel-level mask with a corresponding category for all the instances.

We use the COCO detection metrics [47] as the prevailing quantitative evaluation metrics to evaluate the building extraction performance. Therefore, the evaluation of instance segmentation algorithms is mainly based on the intersection over union (IoU) of predicted mask and ground truth. We particularly use AP_{mask}, i.e., the average precision of mask IoU between references and predictions, in COCO metrics.

The AP is calculated from the precision–recall curve (PRC). Specifically, a prediction is identified as true positive (TP) if the mask IoU between itself and the ground truth is great than a threshold S (e.g., 0.5). False positives (FPs) are falsely predicted objects, while false negatives (FNs) are missing objects. Denoting TP, FP, and FN as the numbers of TP, FP, and FN samples, precision and recall are defined as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP}+\text{FP}} \tag{3}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP}+\text{FN}}. \tag{4}$$

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TABLE III
CLASSWISE INSTANCE SEGMENTATION RESULTS USING AP (%) ON UBCV2 TEST SET WITH FINE-GRAINED ROOF CLASSES: FL-FLAT, GA-FABLE, GM-GAMBREL, RO-ROW, ME-MULTIPLE EAVE, H1-HIPPED V1, H2-HIPPED V2, MA-MANSARD, PY-PYRAMID, AR-ARCHED, RE-REVOLVED, AND OT-OTHER

Method	AP	AP_{50}	FL	GA	GM	RO	ME	H1	H2	MA	PM	AR	RE	OT
SOLOv2 [30]	14.2	24.0	24.6	21.3	27.7	5.8	9.5	7.6	35.0	3.6	4.6	14.8	**11.2**	6.1
QueryInst [53]	15.3	25.2	25.5	23.4	28.6	5.1	24.2	8.9	33.9	7.0	4.8	13.9	4.5	5.5
Mask R-CNN [32]	15.5	25.9	25.8	24.7	**29.2**	5.1	12.3	10.0	39.0	5.4	5.6	16.8	5.8	**7.2**
Cascade Mask R-CNN [34]	16.5	26.9	26.5	24.5	29.1	4.7	19.7	10.9	38.6	6.0	**7.0**	**18.5**	6.7	6.4
Mask R-CNN + **CGT**	16.3	26.3	25.2	**25.1**	27.4	**9.0**	21.6	9.8	39.6	5.7	5.8	17.2	3.2	6.8
Cascade Mask R-CNN + **CGT**	**17.1**	**27.1**	**27.2**	25.0	28.6	6.6	**24.9**	**11.1**	**39.8**	**7.7**	5.8	16.4	5.7	**7.2**

To evaluate precision and recall simultaneously, AP is used to compute the area under the PRC. For the multicategory case, mAP is used to represent the mean AP of all the categories calculated as

$$\text{mAP} = \frac{1}{n}\sum_{i=1}^{n} \text{AP}_i \tag{5}$$

where n denotes the number of categories, and i is the index of the class.

B. Implementation Details

In this section, we first introduce the implementation details of our experiments. Then, we introduce in three sections: 1) the baseline experiments with the state-of-the-art models on fine-grained roof instance segmentation; 2) the experiments for transfer between different styles of cities; and 3) the experiments on multimodal images.

We use the MMDetection [54] tools in our implementation. For all the experiments, we train on 4 × 2080Ti GPUs for 36 epochs with one image per GPU. During training, we evaluate the model on the validation set at the end of every epoch. After 36 epochs of training, we find that the performance on the validation set does not increase more. For different models and modalities, we use different learning rate tuning strategies, tuning each model to the optimum. During training, multiscale and random flipping are used as data augmentation, with images randomly rescaled between 512 × 512 and 896 × 896 pixels. During testing, all the images are rescaled to 768 pixels. The maximum number of detections is set to 300 for all the models. To ensure fairness, the remaining hyperparameters of the models follow the default MMDetection setup.

C. Fine-Grained Roof Instance Segmentation

We conduct our baseline experiments using four popular instance segmentation models: Mask R-CNN [32], Cascade Mask R-CNN [34], SOLOv2 [30], and QueryInst [53]. Mask R-CNN is one of the most classic and still one of the most powerful two-stage algorithms. It adopts a mask-head branch to regress the masks of objects in parallel to the bounding box branch. Cascade Mask R-CNN combines the ideas of Mask R-CNN and Cascade R-CNN [34] to generate more accurate mask predictions on the basis of the cascade paradigm. SOLOv2 is an efficient one-stage algorithm for instance segmentation. We adjust the grid size of SOLOv2 [30] to 60, 54, 36, 24, and 18 due to the dense buildings with smaller sizes in the dataset. And we use Dice loss and focal loss as losses for the mask and classification branches, with weights of 3 and 1, respectively. QueryInst improves the performance of end-to-end instance segmentation by introducing query-based methods and attention mechanisms. We implement the above baseline models with the same backbone network. We systematically compare and analyze the performance of these advanced instance segmentation algorithms on our dataset. Our backbone network for all the models is ResNet-50 [48] with FPN [55] as the neck network.

Quantitative results presented in Table III reveal that the two-stage methods [32], [34] usually outperform the single-stage methods [30], [53]. Mask R-CNN [32] achieves 15.5% AP and 25.9% AP_{50}. The improved Cascade Mask R-CNN [34] model outperforms it by 1.0% in both the metrics. The cascade detection scheme helps improve the localization accuracy of buildings. SOLOv2 [30] achieves the worst performance as the single-stage methods cannot localize the densely distributed buildings in UBCv2, which can be seen in the second column in Fig. 15.

The proposed CGT module contributes to the performance improvement in the fine-grained roof instance segmentation task. As can be seen from Table III, after adding the CGT module, the AP of Mask R-CNN [32] and Cascade Mask R-CNN [34] improves by 0.8% and 0.6%, respectively. Cascade Mask R-CNN with CGT achieves the best performance in six categories, indicating that the proposed CGT module improves the geometric modeling of buildings. To further verify the robustness of our CGT module, we run a numerical stability test with different random seeds. Specifically, we conduct five similar experiments, where the only difference is that the random seeds are set from 1 to 5 respectively. In five experiments, the variance of mAP and AP_{50} is 0.16% and 0.13%, respectively. It demonstrates the robustness of the CGT module.

Qualitative visualization results are presented in Fig. 15. It can be found that dense clusters of small buildings (as shown in the rectangle of the first row in Fig. 15) present a challenge for the popular instance segmentation algorithms such as Mask R-CNN [32]. For the example in the first row, QuerInst [53] has a higher recall, yet it misses the large object in the fifth row. This indicates that QueryInst can better deal with a dense distribution of buildings than the other algorithms. The SOLOv2 [30] predicts more FPs compared with other baseline models. And for the examples in the last row, Mask R-CNN

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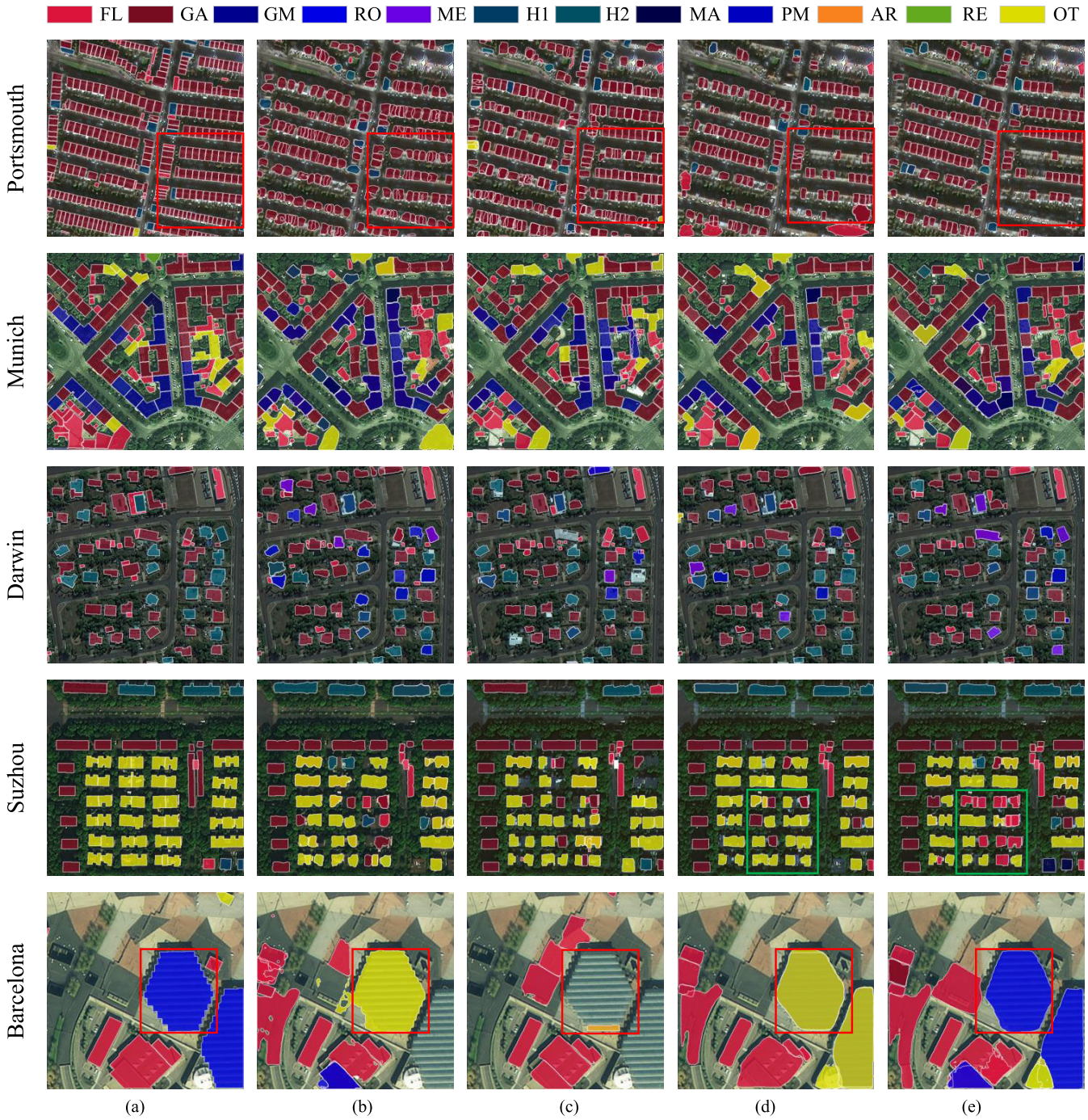


Fig. 15. Results of fine-grained roof type classification with selected baseline models. (a) Ground truth. (b) SOLOv2. (c) QueryInst. (d) Mask R-CNN. (e) Mask R-CNN + CGT.

successfully predicts all the large buildings. It can be found that the addition of the CGT module to Mask R-CNN derives stronger performance for classes with a distinct geometrical structural characteristic such as row roofs.

D. Diversity and Generalization

To verity the importance of the impact of diversity on generalization, we conduct experiments on two cities with large differences in architectural styles, i.e., Beijing and Berlin. Beijing's architecture has a distinct Chinese style, while Berlin's architecture follows a typical European style. We train two instance segmentation models on the data from Beijing and Berlin areas, respectively (the data split is consistent with the globalized version), and then train a global instance segmentation model on the data from all 20 cities. We choose Mask R-CNN [32] as the baseline model for our experiments. Finally, we use the three trained models based on the global, Berlin, and Beijing datasets for inference on the test set images of Beijing and Berlin to obtain the cross-validation results.

Quantitative results presented in Table V demonstrate the importance of diversity for the generalization. It shows that the AP of the model achieves nearly 65% when training

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TABLE IV
INSTANCE SEGMENTATION RESULTS USING AP (%) ON UBCV2 MULTIMODAL TEST SET

Modality	Pretrained	AP	AP_{50}	FL	GA	GM	RO	ME	H1	H2	MA	PM	AR	RE	OT
RGB	Y	**13.0**	**22.9**	**22.9**	**23.5**	**28.8**	**5.0**	**12.3**	**4.9**	**35.9**	**1.2**	**5.7**	**4.8**	**9.3**	**3.3**
RGB	N	3.5	7.6	8.3	9.4	10.5	0.2	3.4	0.3	8.4	0.2	0.1	0.3	0.0	1.0
RGB+SAR	Y	7.9	15.5	16.8	15.2	19.4	3.5	5.5	1.2	15.6	0.7	2.9	1.0	12.2	1.8
RGB+SAR	N	2.3	5.4	5.5	7.3	5.3	0.2	1.8	0.2	6.4	0.0	0.1	0.2	0.0	0.6

TABLE V
CROSS-VALIDATION USING AP (%) ON MODELS TRAINED WITH DATA FROM DIFFERENT CITIES

Train	Test	AP	AP_{50}
Beijing	Beijing	38.4	64.1
Berlin	Beijing	12.3	27.3
Global	Beijing	**40.1**	**65.6**
Berlin	Berlin	36.3	66.4
Beijing	Berlin	15.0	33.4
Global	Berlin	**39.5**	**69.6**



Fig. 16. Cross-validation of models with different training/test datasets (Beijing, Berlin, and global).

and validating on the data from a single city. However, it drops dramatically (more than 30% of the AP metric) when training and validating on different cities. This indicates that it is difficult to transfer the model between cities. Furthermore, the best performance is achieved by the model trained on the global data. Fig. 16 presents results for the three models for Beijing (first row) and Berlin (second row), respectively. The model trained on the global data (third column) produces better results than models trained on data from a single city (the first two columns). It further demonstrates the significance of a dataset covering global cities.

E. Multimodal Baselines

In our experiments, we first conduct a baseline experiment on the RGB-only data yet limited to the 17 cities for which aligned SAR data are available. For the fusion of the two modalities, we directly concatenate the aligned RGB and SAR data in the channel dimension to obtain a four-channel image. This is input to ResNet [48] whose input channel in the first block is adapted to 4.

We choose Mask R-CNN [32] for our experiments. We conduct our experiments in two kinds of settings: training from scratch and fine-tuning the ImageNet pretrained weights. For RGB-only images, we use the weights pretrained on ImageNet to initialize the parameters of the model. For multimodal images, there is a misalignment between the input of the fused multimodal data (four channels) and the pretrained weights (three channels). Therefore, we randomly initialize the first block in the backbone using the strategy proposed in [56].

The results in terms of AP on the multimodal test set of UBCv2 are listed in Table IV. Fine-tuning the ImageNet pretrained weights (second and fourth rows of Table IV) gives better results compared with those trained from scratch. The AP score of the multimodal results is 5.1% lower than that of the single RGB modality. This may be due to the parameters of the first block in the backbone being initialized as random parameters instead of pretrained parameters. And the simple concatenation in the channel dimension affects the representation of the RGB features. A better fusion strategy instead of simple concatenation could improve the performance of multimodal models, e.g., adopt two backbones with shared parameters and fuse multimodal features extracted by backbones. For training from scratch, the model shows the same performance for some types (row roof and pyramid roof) on RGB and multimodal data. This may be because these two roof types exhibit similar visual characteristics in RGB and SAR data. It is unfortunate that the simple early concatenation fails to improve the result. We leave it to future work on how to use multimodal data.

VI. CONCLUSION

In this article, we propose a novel multimodal remote sensing benchmark for buildings with a focus on individual building detection and fine-grained classification from satellite images. The dataset provides annotations for 499 339 building instances in total, and each building is carefully labeled with polygons and a single category. We categorize buildings into 12 fine-grained classes according to the roof types. To guarantee the diversity of buildings in terms of appearance and spatial arrangement, we select 20 cities from five continents, which provides a foundation for the building detection over the world. For all 20 cities, optical satellite imagery is available and for 17 of them additional multimodal imagery with aligned optical and SAR pairs. Along with the selected state-of-the-art baseline models, we propose an CGT module to learn the geometric information of building roofs. With the additional usage of the CGT module, the Mask R-CNN and Cascade Mask R-CNN improve by 0.8% and 0.6% on AP, respectively.

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The experiments demonstrate the great challenges of the benchmark and the improvement using the CGT module.

For future work, we believe that the usage of the multimodal data is worth study, despite that it is not fully explored in this work due to our limited energies. Even though the two kinds of multimodal data are finely aligned, the roof of a specific building always appears at different locations in the optical and SAR images because of the disparities of the imaging mechanisms and conditions. It poses extra difficulties in the usage of the SAR data. A further exploration into this problem and finding an effective solution will be challenging but meaningful as well, which is expected to improve the accuracy of both detection and fine-grained classification.

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